

--	--	--	--	--	--	--	--

B.Tech. Degree VII Semester Regular/Supplementary Examination November 2023

ME 19-205-0705 SUPPLY CHAIN MANAGEMENT (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Define a supply chain and understand its different structures and its importance to the success of a firm.
- CO2: Understand the concept of strategic fit between the SC strategy and the competitive strategy of the firm and how to achieve it.
- CO3: Identify the main drivers of supply chain (SC) performance and measure the musing precise metrics.
- CO4: Assess the importance of distribution networks and the different opinions available for their design.
- CO5: Develop a methodological framework for network design and facility location and use optimisation models and techniques for facility location and capacity allocation.
- CO6: Use safety inventory to deal with demand uncertainty in the supply chain.
- CO7: Understand the importance of transportation in the SC, identify the different modes of transportation and the role played by the infrastructure and policies.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

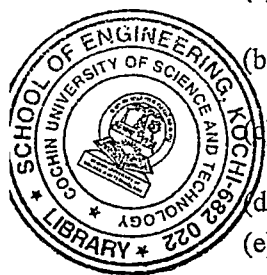
(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Discuss the goal of a supply chain and explain the impact of supply chain decisions on the success of a firm.	3	L1	1	1
(b) Discuss the importance of expanding the scope of strategic fit across the supply chain.	3	L2	2	1
(c) Identify and explain the role of any two major drivers of supply chain performance.	3	L2	3	1
(d) What is the relationship between service levels and inventory levels?	3	L3	6	1
(e) Compare and contrast issues involved in vehicle routing in a courier company.	3	L3	4	2
(f) What are the difficulties in implementing cross docking?	3	L2	7	1
(g) How is managing a multi-planned international network different from managing a domestic multi-planned network?	3	L3	5	1
(h) Why should global firms question their network design decisions every few years?	3	L2	5	1

PART B

(4 × 12 = 48)

- II. (a) Identify the 3 key supply chain decision phases and explain the significance of each one. 4 L2 1 1
- (b) Describe the push/pull views of a supply chain. 4 L1 1 1
- (c) Explain why achieving strategic fit is critical to a company's overall success. 4 L3 2 2

OR**(P.T.O.)**

BTS-VII(R/S)-11-23-1970

		Marks	BL	CO	PO
III.	(a) Consider the supply chain involved when customer purchases a book at a bookstore. Identify the cycles in this supply chain at the location of the push/pull boundary.	4	L2	1	1
	(b) How could a bicycle manufacturer increase responsiveness through its facilities?	4	L3	4	2
	(c) Why does the inventory in a system increase with an increase in the stock points in the system?	4	L2	3	2
IV.	(a) What factors should a manager consider while deciding service levels?	3	L2	5	1
	(b) While computing the ordering cost, one is supposed to consider only the fixed component of the ordering cost. Why?	4	L2	5	2
	(c) A regional warehouse purchases hand tools from various suppliers and then distributes them on demand to retailers in the region. The warehouse operates 5 days a week, 52 weeks per year. The following data are estimated for a 1 inch drill bit.	5	L3	6	2
	Average daily demand	= 100			
	Standard deviation of daily demand	= 30			
	Supplier lead time	= 3 days			
	Holding cost	= ₹9.4 per unit per year			
	Ordering cost	= ₹35 per order			
	Service level	= 98%			
	Design an inventory control system for this project.				
	OR				
V.	(a) Apollo hospital orders its antibiotics every 4 weeks when a sales person from a pharmaceutical company visits it. Zombacycline, which costs ₹25 per capsule is one of its most prescribed antibiotics, with an average daily demand of 50 capsules. The standard deviation of daily demand derived from examining prescriptions filled over the past 6 months found to be 15 capsules. It takes 2 weeks for an order to arrive. The hospital would like 99% of all demands from prescriptions to be satisfied from stock. The cost to place an order is ₹1000 and the holding cost is 20% of the purchase price. The sales person has just arrived and there are currently 1000 capsules in stock. How many capsules should be ordered?	7	L3	6	2
	(b) What is the impact of inventory centralization on various supply chain performance measures?	5	L2	5	1
VI.	(a) Compare and contrast issues involved in vehicle routing in an employee pickup for a software company.	4	L2	4	1
	(b) How do transport companies benefit from a technology like global positioning system (GPS)?	4	L2	7	1
	(c) What are the difficulties in implementing cross docking?	4	L1	7	1
	OR				
VII.	(a) Indian railways has a common pool of marketing officers who service all its clients to explore the idea of creating a few industry verticals within marketing. It wants to create specific industry verticals only for those industries that will have significant business potential. Identify a few strategic industries where Indian railways has high business potential, suggest a methodology that can help the railways in identifying the right industry verticals.	6	L3	5	2
	(b) Over a period of time Amazon has built new warehouses located at geographically different parts of the US. Why should an E-retailer need multiple warehouses located at different parts of the country?	6	L3	5	2

(Continued)

BTS-VII(R/S)-11-23-1970

		Marks	BL	CO	PO
VIII.	A global company has put up a captive facility in India to manage a couple of internal backend processes. The CEO of the local unit is worried about the long term competitiveness of the Indian unit. Labour costs has been increasing at the rate of 15% in the last few years and the CEO is worried that in the near future this processes may get shifted to eastern Europe or some other part of the world. What should the CEO do so that the local unit can survive in the long run?	12	L4	5	4
OR					
IX.	Reliance is trying to design its network for its retail operations in Maharashtra from scratch. Suggest a suitable approach.	12	L4	5	4

Bloom's Taxonomy Levels

L1 = 9.2%, L2 = 36.6%, L3 = 34.2%, L4 = 20%.

